L#01: Systems of Units, Charge, Current, Voltage

 # Six basic SI units – International system of unit

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilo gram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd

Derived units in circuit analysis

Quantity	Unit	Symbol
Electrical charge	coulomb	C
Electric potential	volt	V
Resistance	ohm	Ω
Conductance	siemens	S
Inductance	henry	H
Capacitance	farad	F
Frequency	Hertz	Hz
Force	newton	N
Energy, work	joule	J
Power	watt	W
Magnetic flux	weber	Wb
Magnetic flux density	tesla	T

Decimal multiples and submultiples

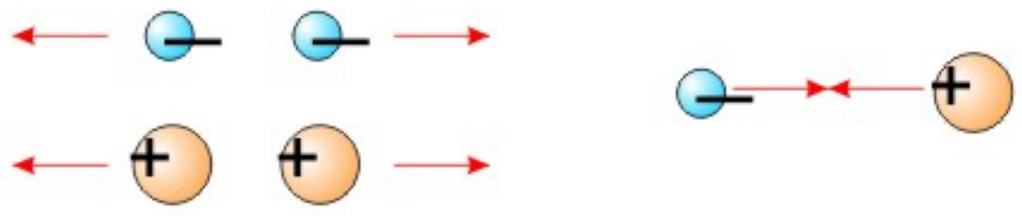
Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

10^{-15}

10^{-18}

⋮

Electric charge

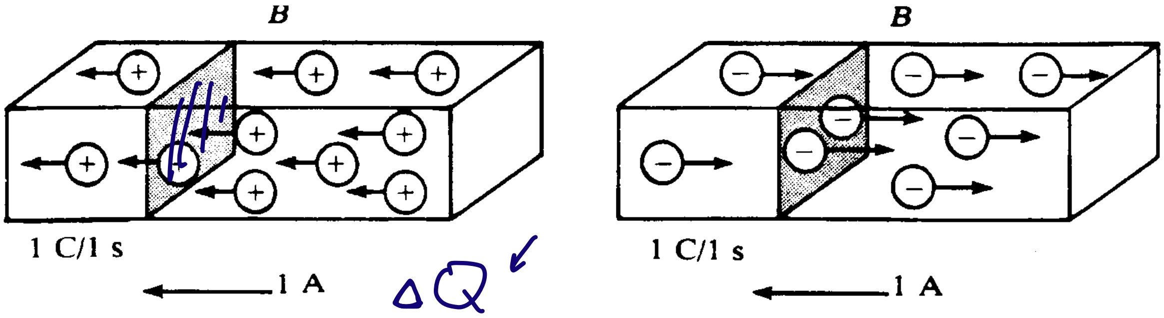


Like Charges Repel   Unlike Charges Attract

- **Charge** is an electric property of the atomic particles, measured in Coulomb, C.
- The charge e on one electron is negative and equal in magnitude to 1.602×10^{-19} C.
- $1C = 1/e = 6.24 \times 10^{18}$ protons.
- **Law of conservation of Charge:** Charge can neither be created nor destroyed, it can only be transferred.

Current

- **Electrical Current:** motion of charges creates electric current.
- **Convention:** Current flows in the direction of positive charges.



(a) $I = \frac{\Delta Q}{\Delta T}$ (b)

- Current is measured in ampere, A, which is C/s.
- Formula:

$\Delta T \to 0.$

ΔT, 1 week.

P1: 3C

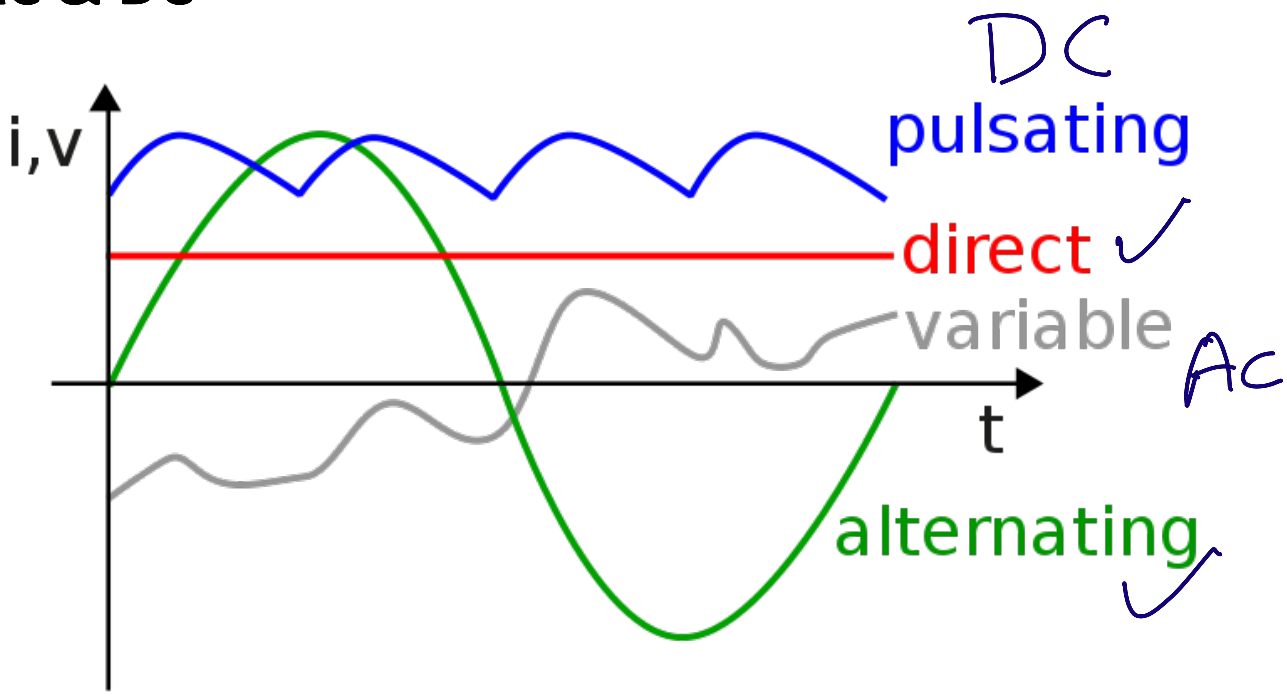
P2: 2.8C.

⋮

$$\boldsymbol{i} =$$

$$i(t) = \lim_{\Delta T \to 0} \frac{\Delta Q}{\Delta T} = \frac{dq}{dt}.$$

AC & DC



Source: https://en.wikipedia.org/wiki/Direct_current#/media/File:Types_of_current.svg

Direct current (DC): the electric current flows in a constont direction.

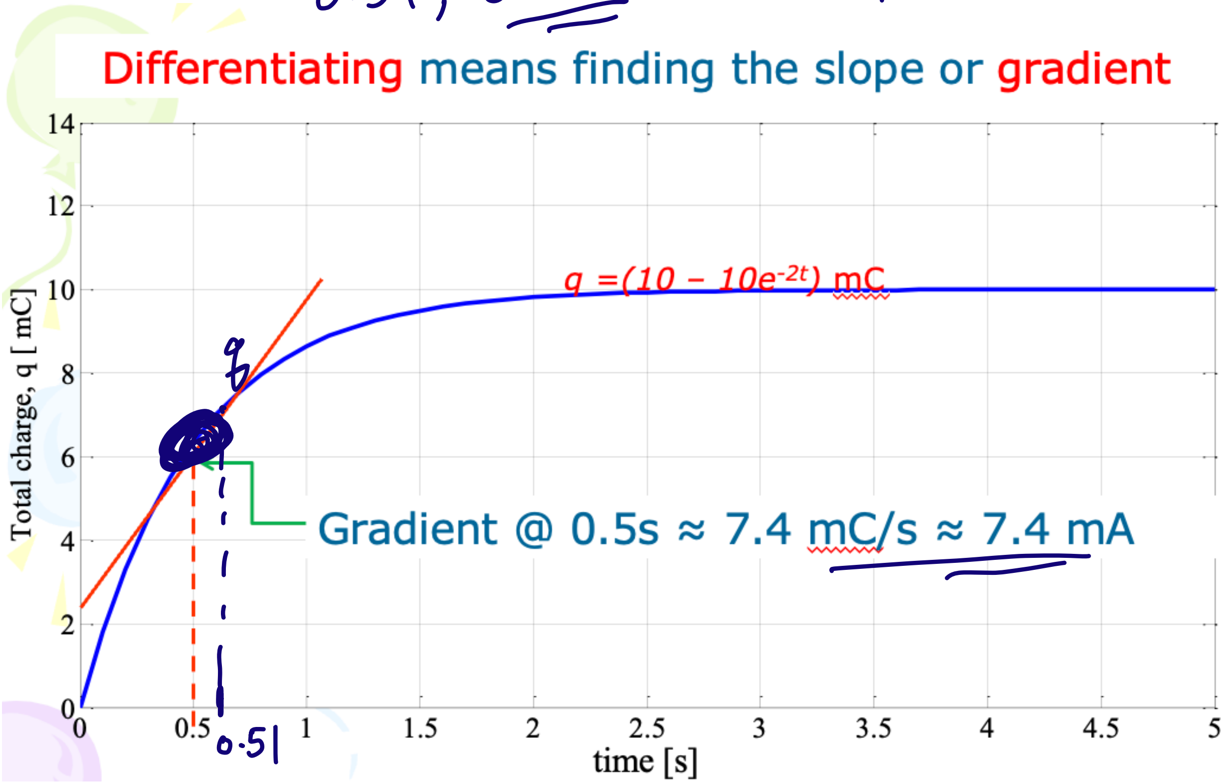
Alternating current (AC): the electric current periodically reverses its direction.

Q1: Calculate the amount of charge represented by four million protons.

$$Q = 4 \times 10^{6} \times 1.602 \times 10^{-19} C.$$

Q2: The total charge entering a terminal is given by $q = (10 - 10e^{-2t})$ mC. Find the **current** at $t = 0.5s$.

0.51, 0.501 $\Delta T \to 0$.



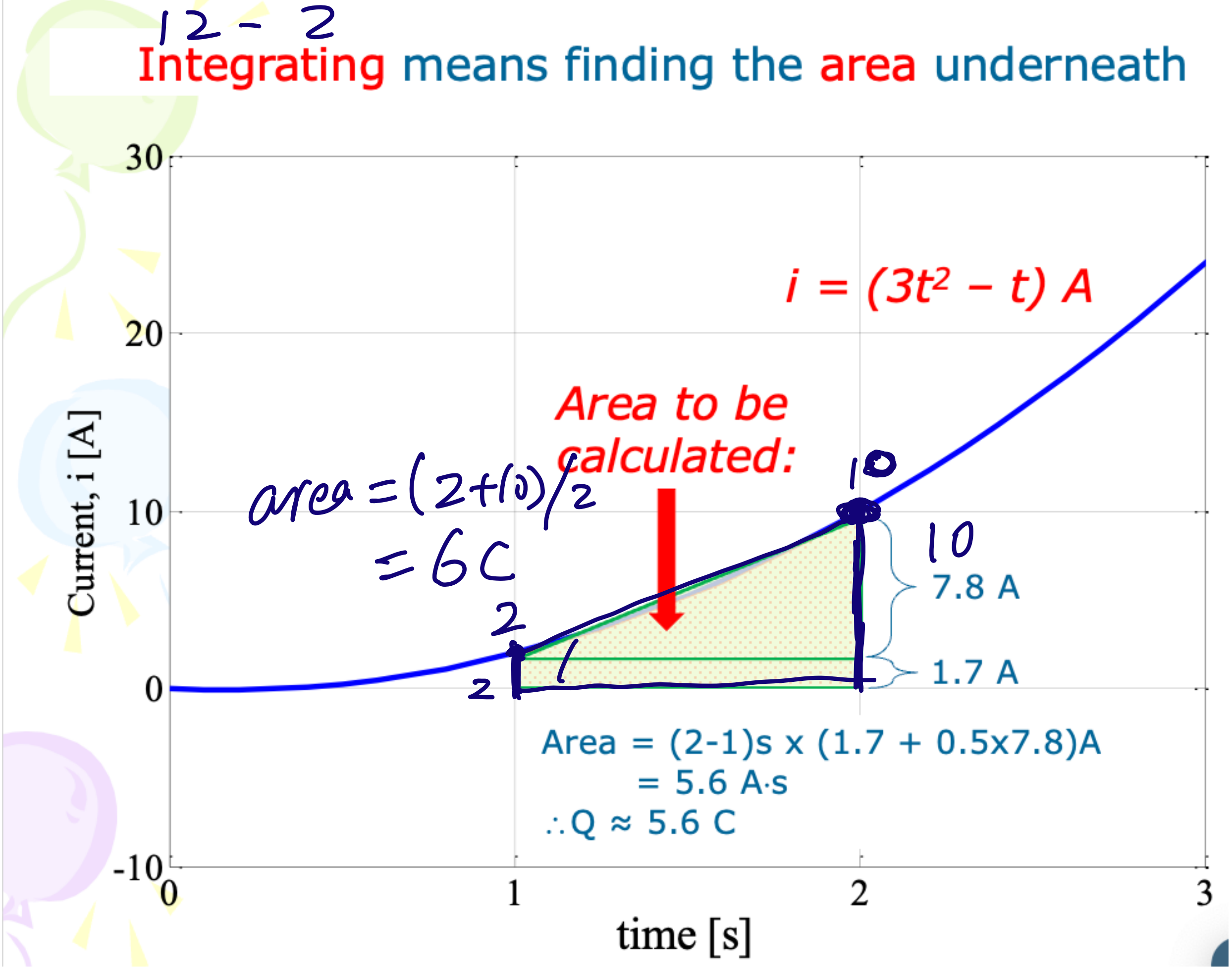
$$I = \frac{\Delta Q}{\Delta T} = \frac{q(0.51) - q(0.5)}{0.51 - 0.5} =$$

$\Delta T = 0.01$,

$\Delta T = 0.001, \ldots$

$I = \frac{\Delta Q}{\Delta T} \rightarrow \Delta Q = I \cdot \Delta T.$

Q3: Determine the total charge entering a terminal between t = 1 s and t =2 s if the current passing the terminal is: $i = (3t^2 - t)\,A$



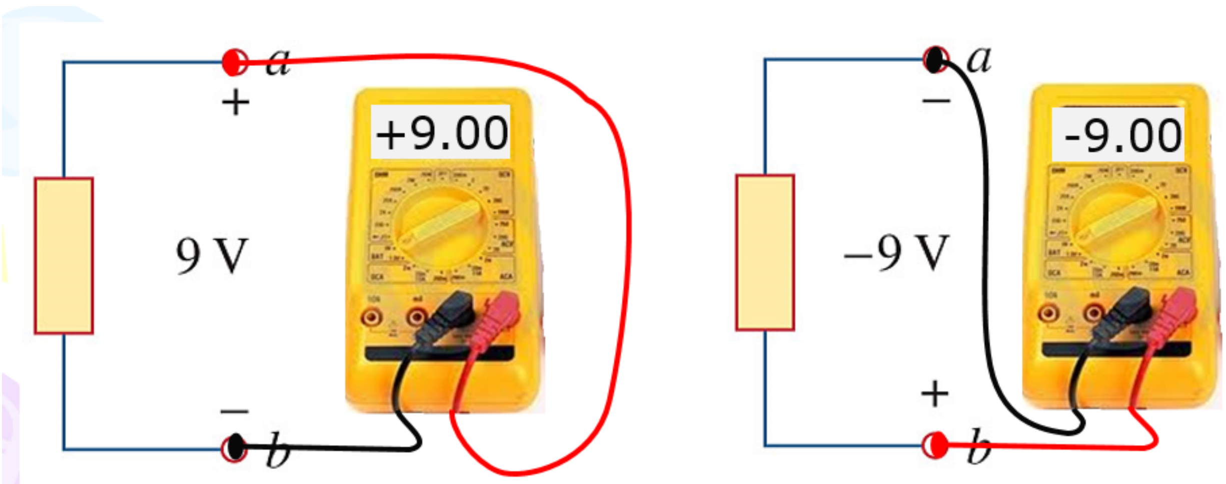
0

Voltage

- Voltage is the potential difference. between two points in an electric field.
- The potential difference is the work done per unit of charge against a static electric field to move a charge between two points, which is measured in volts (a joule per coulomb).

$$V_{ab} = \frac{dw}{dq} = V_{ao} - V_{bo}$$

Compare V_a, V_b



$V_{ab} = V_a - V_b > 0, \quad V_a > V_b.$

$< 0, \quad V_a < V_b$

$= 0, \quad V_a = V_b$

$= -(V_b - V_a) = -V_{ba}.$

$V_{ab} = -V_{ba}.$